

Drone Payload Structure Design and Analysis

In this project, you are optimizing drone arm design, including geometry and material selection, to maximize payload capacity while ensuring structural integrity under motor-induced loads. You will use MATLAB to perform force balance analysis and finite element modeling to quantitatively support your design solution.

Use this Live Script as a template to get started on your project solution. You may include helper functions at the bottom of this document or as separate files. For a tutorial on how to use Live Scripts, check out this [video](#).

Suggested Tasks

1. Gather your starter assumptions and load the material properties into MATLAB.
2. Propose at least two drone arm designs
3. Perform a thrust-to-weight analysis to evaluate each of your designs across all material options. Your design must meet a minimum payload requirement and safety standards.
4. Perform finite element analysis across all design and material options.
5. Recommend a final design solution.

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Break Down the Problem

In the space below, define the problem statement:

The goal is to design a drone arm that can safely handle motor-induced loads while maximizing payload capacity. Two structurally distinct designs must be created and analyzed using finite element analysis (FEA), thrust-to-weight analysis, and provided drone arm material properties. The designs must meet project-specified minimum payload and thrust-to-weight ratio requirements, as well as a minimum factor of safety (FOS) requirement to ensure safe operation.

List the requirements, constraints, and success criteria for the project:

Requirements:

- Minimum payload requirement: 0.5 kg
- Minimum thrust-to-weight ratio: 2:1
- Minimum FOS: 1.5

Constraints:

- Quadcopter configuration: 4 motors and 4 propellers
- Motor mass: 0.065 kg per motor
- Thrust per motor: 1 kg of thrust
- Maximum total thrust: 4 kg of thrust
- Maximum drone arm mass: 0.125 kg per arm
- Total mass of components excluding drone arms and payload: 1.0 kg

Success Criteria:

1. Drone arm must maximize payload capacity while meeting minimum thrust-to-weight ratio and minimum FOS.
2. Drone arm must minimize mass.
3. Drone arm must limit maximum stress and displacement.
4. Drone arm should minimize material cost (optional).

What is your proposed solution?

We propose two possible drone arms. The first drone arm design will use a hollow oval cross-section and one drone body attachment points. The second drone arm will use a triangular truss design and contain three drone body attachment points, and the truss design will contain solid outer supports and solid inner web members.

List the steps you will need to take to achieve your proposed solution:

1. Sketch out two drone arms, making sure to include rough dimensions, a motor attachment point, and drone body attachment point(s).
2. Create the 3D models from sketches using CAD software. Export these models as STL for analysis.
3. Analyze both drone arms and provide initial thrust-to-weight ratio results for each drone arm and material combination.
4. Perform FEA on the drone arms and record x-, y-, and z- displacement as well as Von Mises stress. Ensure drone arms meet the minimum FOS.
5. Use an optimization function that minimizes mass and material cost (optional).
6. Display required results. This includes thrust-to-weight and FEA results for each drone arm and material combination.
7. Analyze the FEA data and compare the advantages/disadvantages of each drone arm. Comparison should include payload capacity, arm mass, FOS, Von Mises stress, and displacement differences.
 - a. Optional: Include material cost comparison.
8. Submit our final report which includes a drone arm design recommendation.

What challenges do you face in the process of achieving the solution?

1. Material selection and tradeoff analysis: Each material has different properties that give it distinct advantages and disadvantages. The challenge arises from selecting a material with advantageous properties that align with the drone arm requirements while minimizing disadvantageous properties.
2. Drone arm design and CAD: The dimensions of the drone arm affect mass, stress, and displacement, which makes dimension selection a vital part of the drone design process. Additionally, the selected dimensions must be realistic to ensure that the drone arm can be manufactured. Due to the team's limited experience in design, the selection of proper dimensions is a challenge.
3. FEA setup: The team members have limited experience with FEA. As a result, properly setting up FEA in MATLAB and accurately interpreting results becomes a challenge.

Task 1: Gather your starter assumptions and load material properties

Load the provided `droneArmMaterials.mat` file into MATLAB. This contains a data struct with the material properties for six different materials commonly used for drones (carbon fiber composite, aluminum alloy, fiberglass composite, PLA plastic, ABS plastic, and wood). For each material, you have been provided the following information:

- Density (ρ), in $\frac{\text{kg}}{\text{m}^3}$
- Young's Modulus (E) in Pa
- Poisson's ratio (ν), unitless
- Yield strength, in Pa
- Average cost per meter in US dollars

For this project, you can assume the following design parameters:

- Quadcopter drone with 4 motors and 4 propellers
- Each motor can provide up to 1kg thrust
- Acceleration due to gravity: $g = 9.81 \frac{\text{m}}{\text{s}^2}$
- Total weight of the drone components without the arms or the payload: 1kg

The breakdown of the weight of each drone component is as follow:

- Battery = 0.3kg
- Each motor = 65g, 4 motors = 0.26kg
- Each propeller = 10g, 4 propellers = 0.04 kg
- Each electronic speed controller = 30g, 4 electronic speed controllers = 0.12 kg
- Flight controller and electronics = 0.05kg
- Receiver/GPS = 0.03kg
- Drone frame (center body) excluding arms = 0.2kg

```
clear, clc, close all
addpath('functions');
materials = importdata("data\droneArmMaterials.mat");
```

```

% Assign file paths for drone arm files
filePaths = [
    "cad/TrussArm.step"
    "cad/OvalArm.step"
];
designNames = [
    "Truss Arm"
    "Oval Arm"
];
areas = [[0.00002827, 0.00002827]]; % m^2

% Arrays of material data

[materialNames, density, youngsModulus, poissonsRatio, yieldStrength, cost] =
materialDataExtractor(materials);
nMaterials = numel(materials);

g = 9.81;          % gravitational acceleration [m/s^2]

nMotors = 4;        % number of motors
nProps = 4;         % number of propellers
nArms = 4;          % number of arms

thrustPerMotor_kg = 1;
maxTotalThrust_kg = nMotors * thrustPerMotor_kg;

mBattery = 0.30;    % battery [kg]
mMotors = nMotors * 0.065; % 4 motors, 65 g [kg]
mProps = nProps * 0.010; % 4 propellers, 10 g [kg]
mEsc = 4 * 0.030;    % 4 ESCs, 30 g [kg]
mFlightCtrl = 0.05;  % flight controller + electronics [kg]
mReceiverGPS = 0.03; % receiver / GPS [kg]
mFrame = 0.20;       % center body, excludes arms [kg]

baseMass = mBattery + mMotors + mProps + mEsc + mFlightCtrl + mReceiverGPS + mFrame;

motorMass = 0.065; % [kg]

minPayload = 0.5; % minimum required payload [kg]
minTWR = 2;      % minimum thrust-to-weight ratio
minFOS = 1.5;    % minimum factor of safety

% Net transverse tip load for the Task 3 stress pre-screen [N]
thrustPerMotor_N = thrustPerMotor_kg * g;
motorWeight_N = motorMass * g;
F_tip_N = thrustPerMotor_N - motorWeight_N; % net upward load bending the arm
FOSmin_preFEA = 2.0; % FOS accounting for hand calculation
errors

```

Task 2: Propose at least two drone arm designs

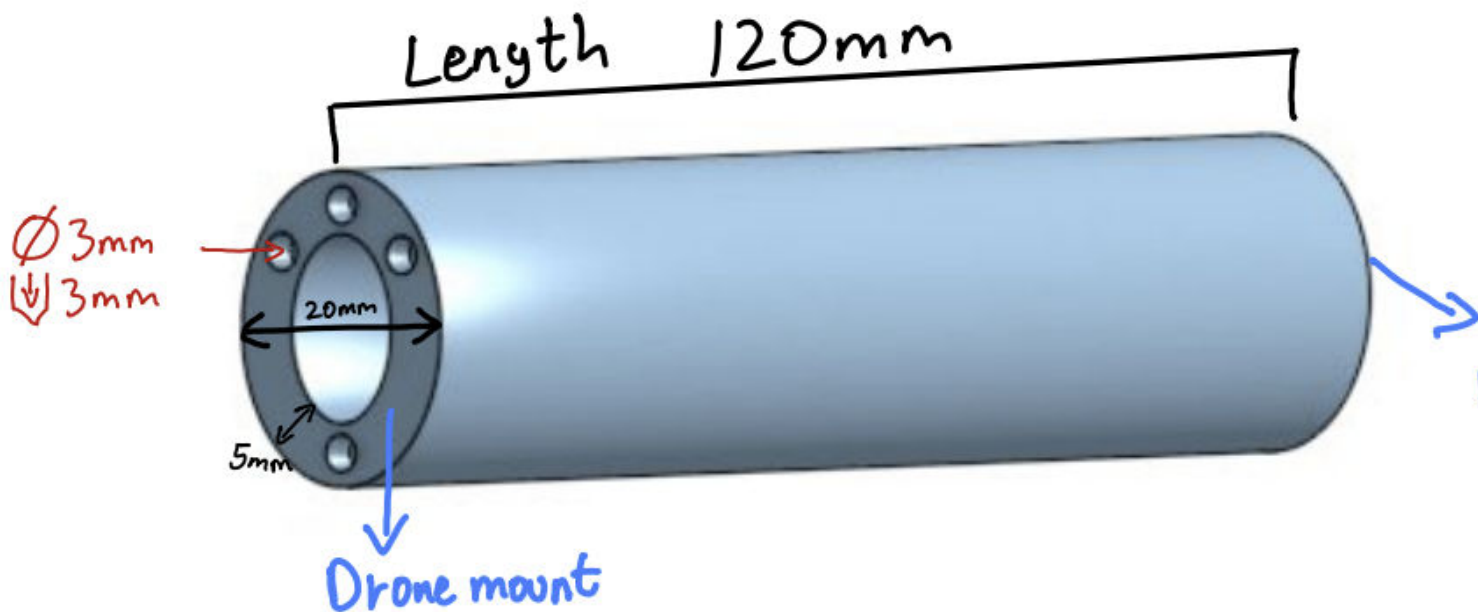
Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features, including:

- Overall shape (e.g. beam, tube, truss structure)
- Dimensions (length, width, thickness)
- Motor mount location
- Where the arm connects to the drone body

The two designs should be meaningfully different from each other (e.g. different geometries or structural strategies) and not just small variations of the same idea.

Navigate to the 'Insert' tab and insert pictures of your sketches in the space below (hand-drawn or digital). Include a short design description for each sketch that includes:

- Geometry (e.g. solid beam, hollow tube, beam with trusses)
- Why you chose this design
- How it might perform under load

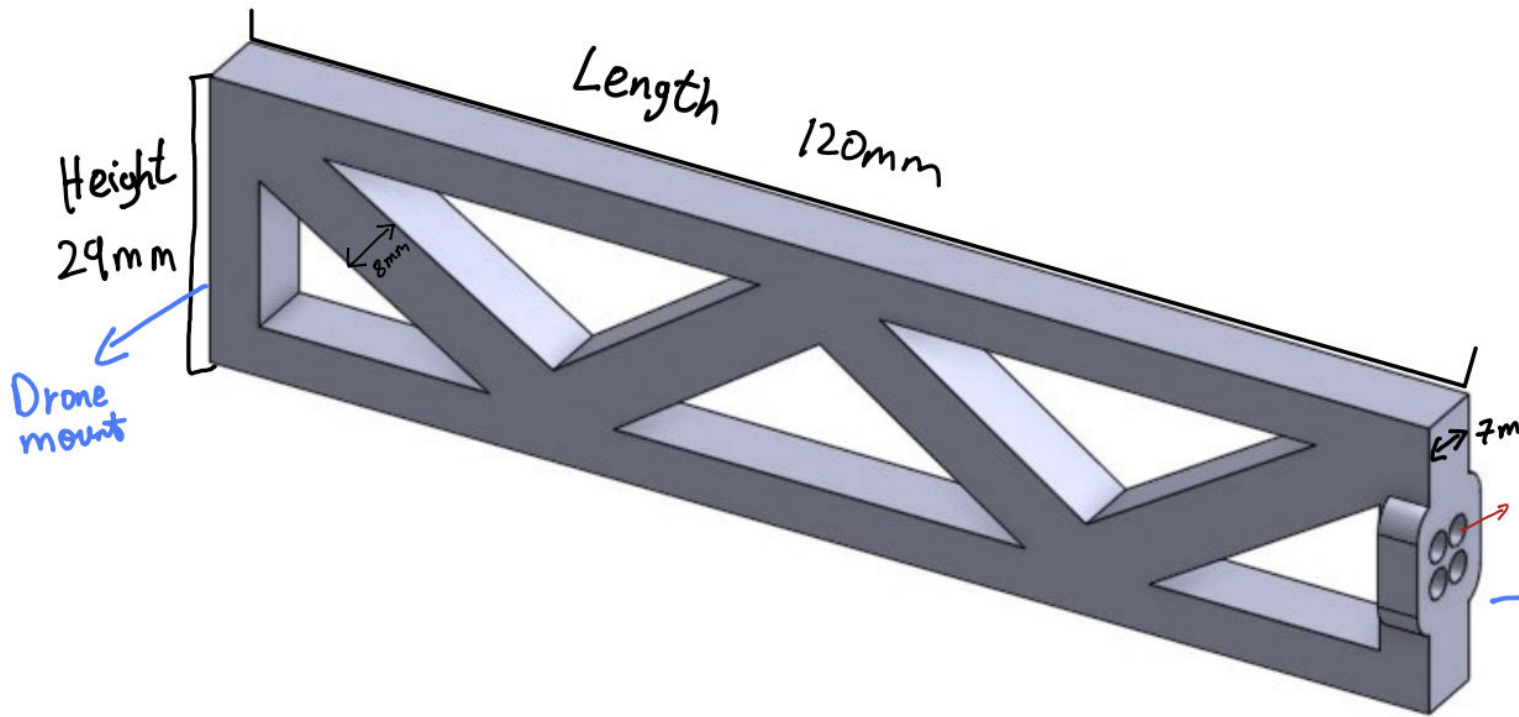


Design 1 — Hollow Oval Tube

Geometry: Hollow oval tube, 120 mm long, 25 mm tall × 20 mm wide outer cross-section with a 5 mm uniform wall (15 × 10 mm inner bore). Motor mounts at the tip via four Ø3 mm × 3 mm holes; the arm bolts to the drone body at the base end through a matching four-hole pattern.

Why we chose it: A closed section distributes bending load around the full perimeter and resists torsion from motor and propeller reaction, which an open section cannot. Hollowing the center removes material near the neutral axis, where bending stress is lowest, so very little strength is given up for the weight saved. The section is taller than it is wide because section depth is cubed in the second moment of area while width contributes only linearly, so the tall orientation resists bending far better at the same mass.

How it performs under load: Acts as a cantilever beam, with stress peaking at the top and bottom outer surfaces where the arm meets the body. Our initial design used a thinner wall, but FEA showed excessive tip deflection, which we judged too flexible for stable flight. Increasing the wall to 5 mm brought deflection down substantially at the cost of additional mass.



Design 2 — Planar Truss

Geometry: Flat truss plate, 120 mm long, 29 mm tall, 7 mm thick, with 8 mm wide members forming triangular bays. Motor mounts at the tip through four $\text{Ø}3 \text{ mm} \times 3 \text{ mm}$ holes in a raised boss; the base end attaches to the drone body across the full-height end face.

Why we chose it: Inspired by bridge trusses, which achieve high strength per unit weight. The triangular pattern removes material from regions carrying little load while keeping material along the top and bottom edges where bending demand is highest, so the arm occupies a large structural envelope while using far less material than a solid beam of the same size.

How it performs under load: Bending is resolved into a tension–compression couple in the top and bottom chords, with the diagonals carrying shear between them. Our initial design used 3 mm members, but FEA showed factor of safety below the 1.5 requirement for several material options, driven by stress concentrations at the joints that our analytical model did not capture. Widening the members to 8 mm resolved this across all six materials.

Task 3: Perform thrust-to-weight analysis across all design and material options

Write a MATLAB function that will perform a thrust-to-weight analysis across all design and material options. This analysis will help you evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements.

- Your design must support a minimum payload of 0.5kg.
- Your design must meet a thrust-to-weight ratio of at least 2:1 to ensure safe and stable flight. This safety standard is widely used in drone design. Conceptually, it means that the drone's maximum thrust must be at least twice its weight. This ensures stable hovering and maneuvering with a safety margin. Note that the thrust-to-weight safety margin is a system-level safety standard for drone flight that is distinct from the structural safety factor of the drone arms (which you will evaluate later through finite element analysis).
- Your goal is calculate maximum payload capacity for each design and material option (i.e. what is the heaviest payload your drone design could carry while still allowing the drone to safely take off?)

Ensure that the results of your analysis can be readily interpreted, e.g. in the form of a summary table or plot.

Note: if you have two drone arm designs, the expected output should be 12 payload capacity estimates (2 designs across 6 materials options).

```
% Calculate the volumes of each design (m^3 * kg/m^3)
volumes = calculateDesignVolume(filePaths);

% calculate the mass for each combination of material and design
armMass = density * volumes; % kg
totalArmMass = armMass .* nArms; % kg

% calculate max allowed mass and total mass
maxAllowedMass = maxTotalThrust_kg / minTWR; % kg
totalMass = totalArmMass + baseMass; % kg

% Calculate max payload
maxPayload = maxAllowedMass - totalMass; % kg

% Check to see if material and design combination meets the requirements
meetsRequirementMinPayload = maxPayload >= minPayload;
```

Task 4: Perform finite element analysis across all design and material options

Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. To simplify the task, you will perform the FEA only on a single drone arm.

Using a CAD software of your choice, first create a 3D model of one drone arm for each of your designs. Be sure your model can be exported as a STEP or STL file for analysis in MATLAB.

- Recommended CAD software options: Tinkercad, Onshape, Fusion 360, Solidworks (most have free versions)

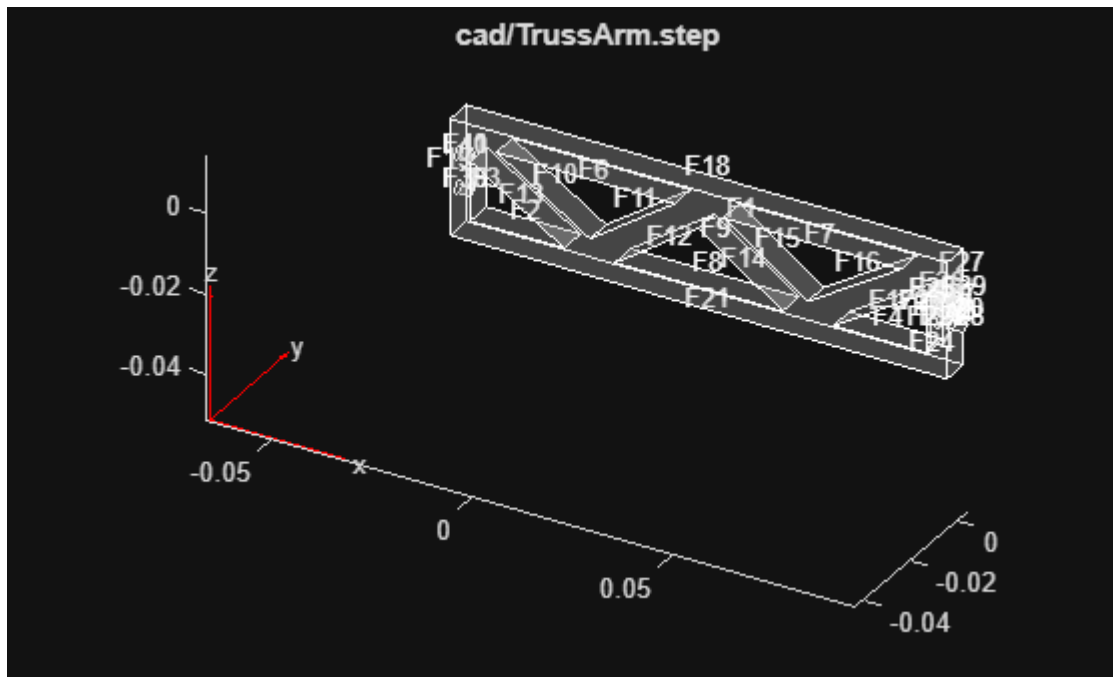
Import your model into MATLAB and perform FEA. In your FEA model, you should model and apply two loads to the drone arm: upward thrust force from the motor and downward weight of the motor.

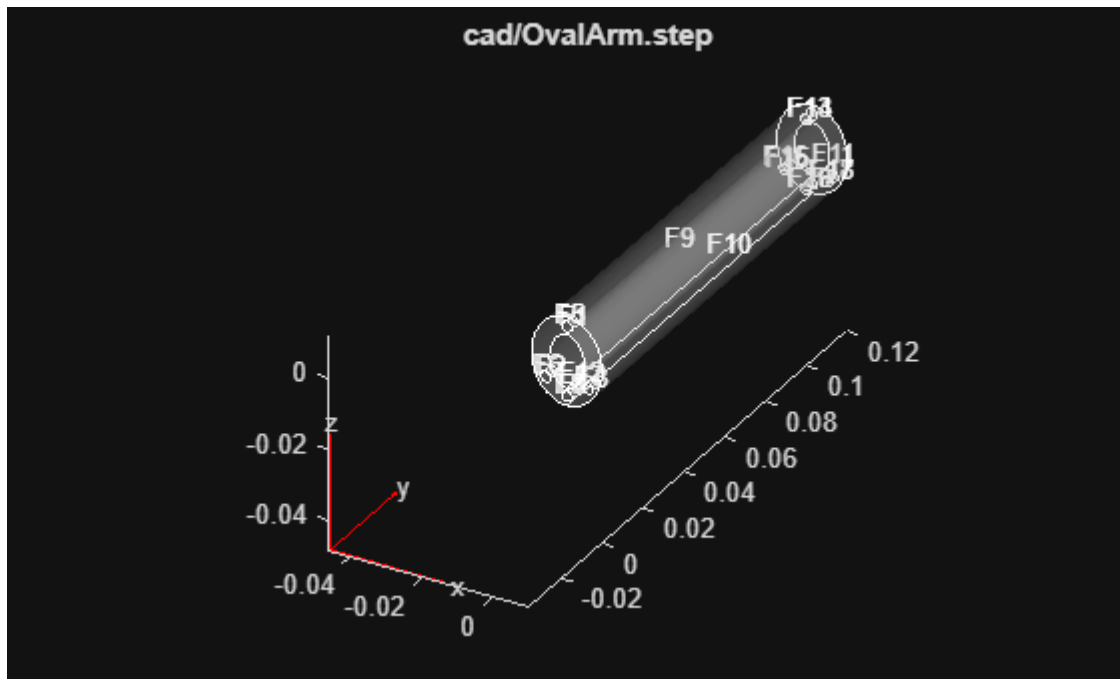
- Helpful resources: [online tutorial](#), documentation for [Structural Mechanics](#) and [Partial Differentiation Equation Toolbox](#), and [example workflow](#)

For each design and material combination, you should report the overall maximum displacement, maximum stress, and factor of safety and include visualizations of the x-, y-, and z-displacement and Von Mises stress. Ensure that your results are presented for easy interpretation (e.g. present a table with numerical results, clearly labeled subplots for the visualizations).

- The factor of safety compares how strong a material is to how much stress it experiences, indicating whether a design is safe under the given loads. For this project use the equation:
$$\text{FoS} = \frac{\text{Material Yield Strength (Pa)}}{\text{Maximum von Mises Stress}}$$
Yield strength indicates how much stress a given material can withstand before permanently deforming. Maximum von Mises stress (derived from your FEA model) is a measure of how much stress your design experiences under the applied loads.
- A typical engineering safety margin for FoS is ~1.5-2+
- This [example](#) demonstrates the plots you are asked to include in your results

```
% Runs design visualizer to determine which faces are fixed and loaded  
designVisualizer(filePath)
```





% Set fixed and loaded faces

```
fixedFaces = {[39, 41] , [5, 6, 7, 8]};
```

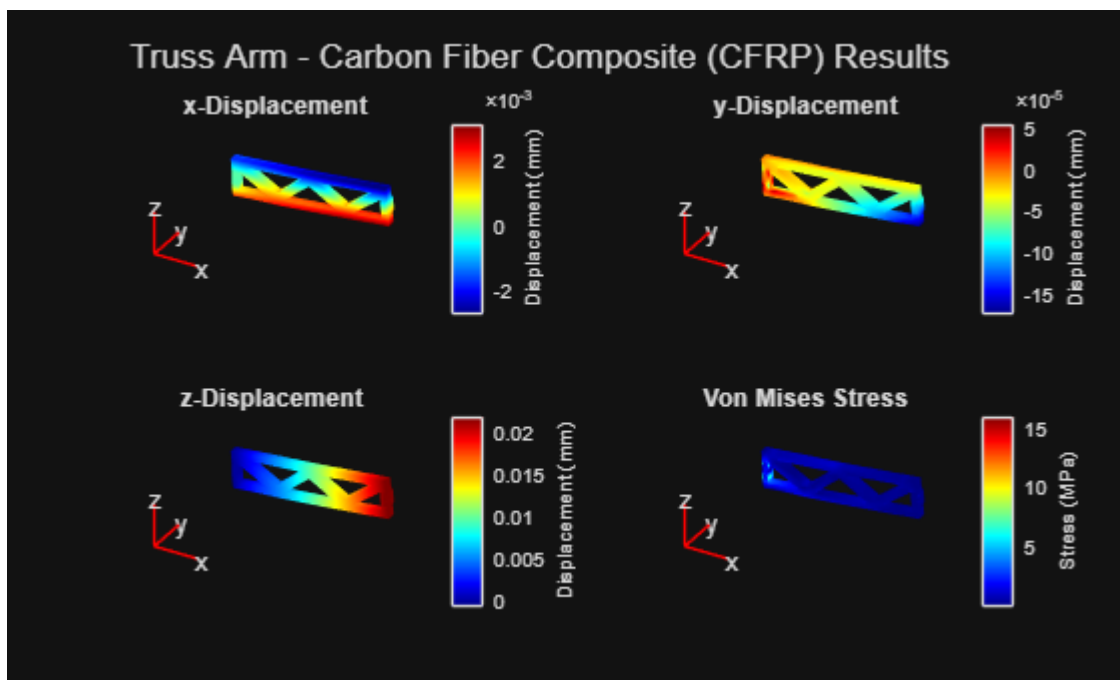
```
loadedFaces = {[31, 33, 35, 37], [14, 16, 18, 20]};
```

% Runs the FEA analysis

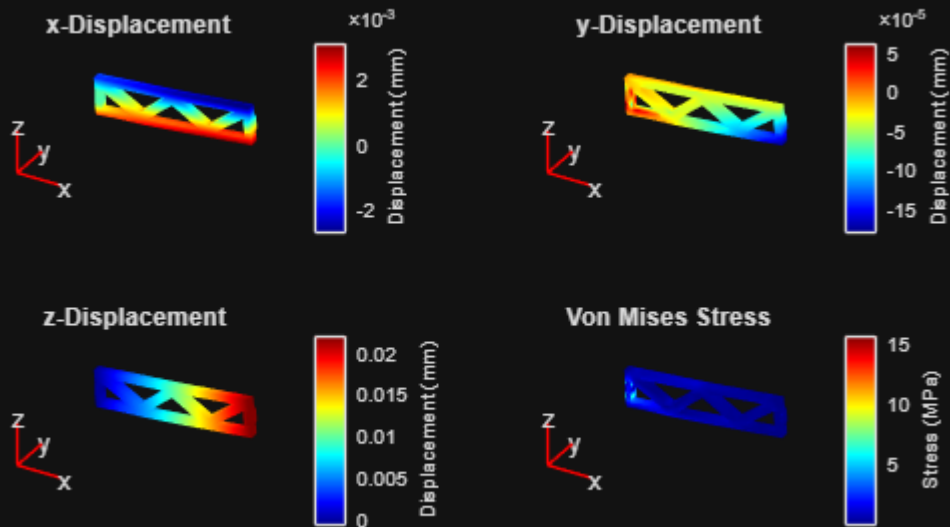
```
results = droneArmFEA(filePaths, areas, fixedFaces, loadedFaces, designNames,  
materialNames, youngsModulus, poissonsRatio, density, yieldStrength);
```

% Generates graphs from FEA analysis

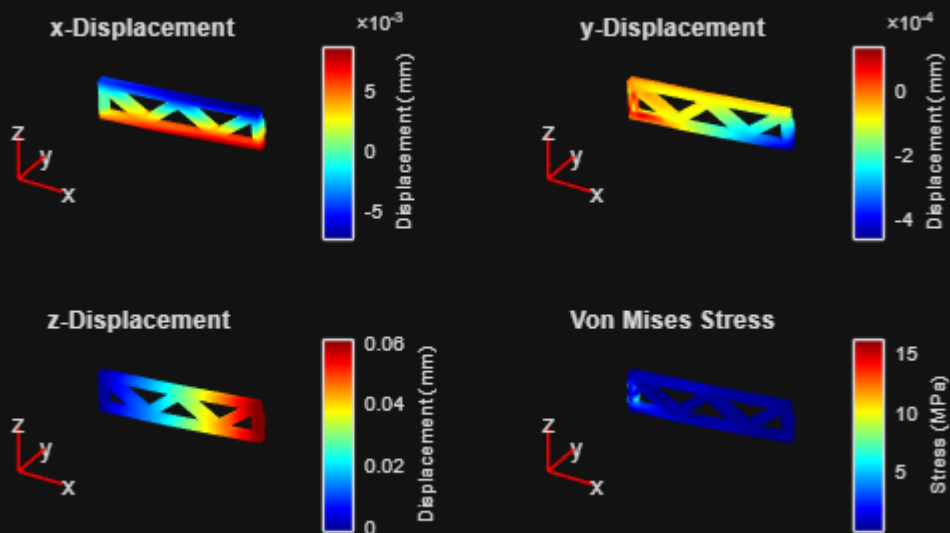
```
generateGraphsFEA(results)
```



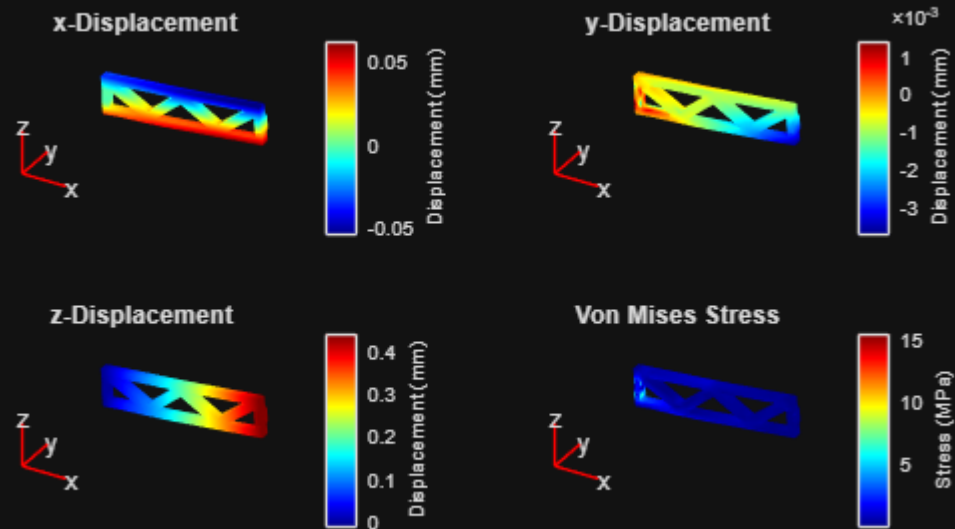
Truss Arm - Aluminum Alloy Results



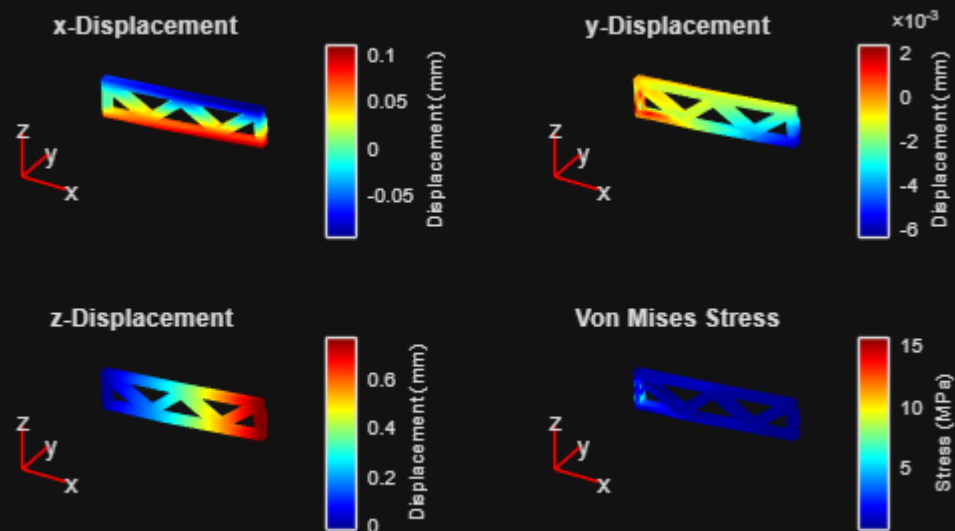
Truss Arm - Fiberglass Composite (GFRP) Results



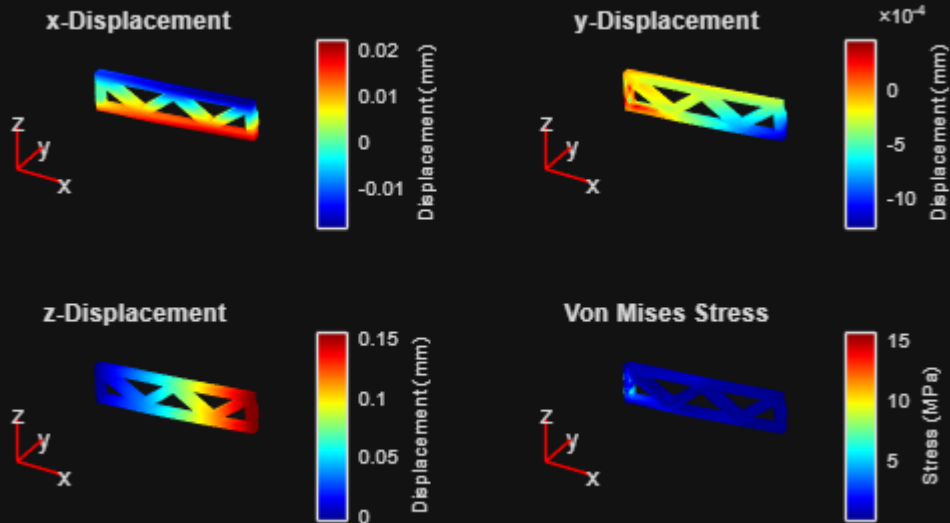
Truss Arm - PLA Plastic Results



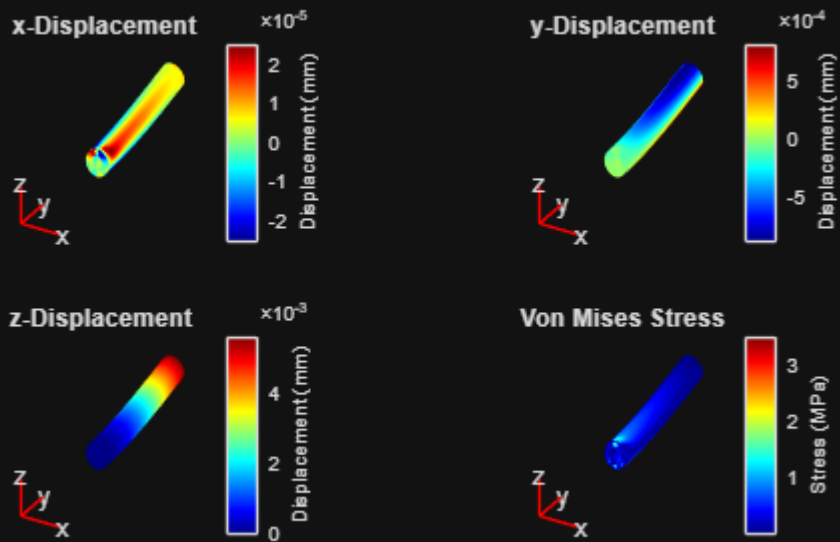
Truss Arm - ABS Plastic Results



Truss Arm - Wood (Birch) Results

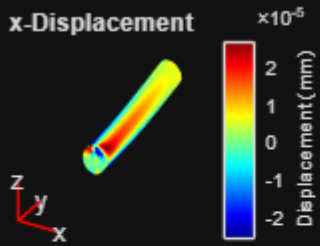


Oval Arm - Carbon Fiber Composite (CFRP) Results

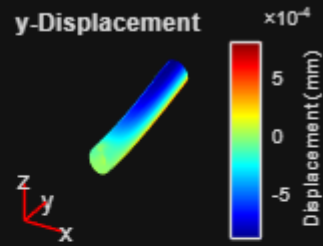


Oval Arm - Aluminum Alloy Results

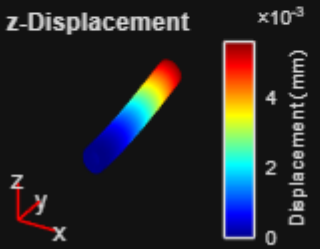
x-Displacement



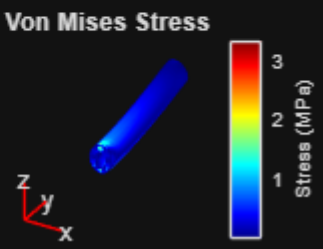
y-Displacement



z-Displacement

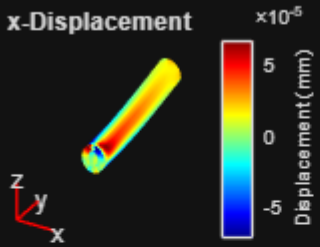


Von Mises Stress

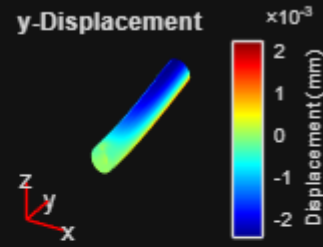


Oval Arm - Fiberglass Composite (GFRP) Results

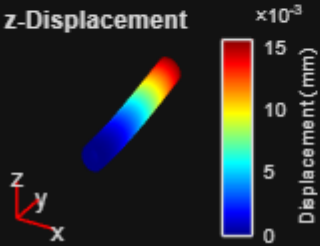
x-Displacement



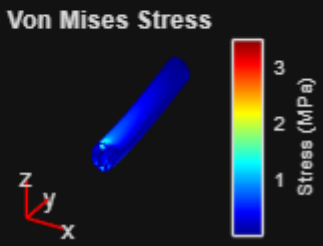
y-Displacement



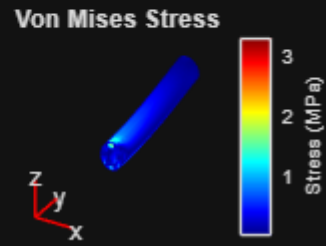
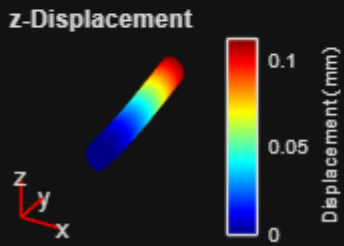
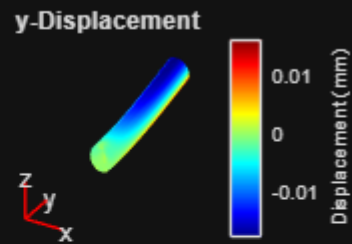
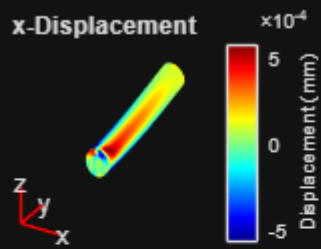
z-Displacement



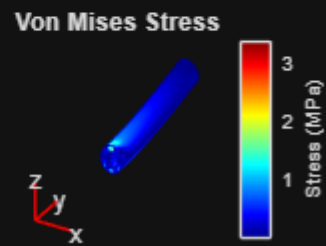
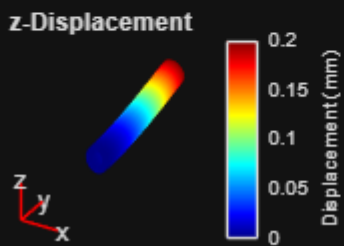
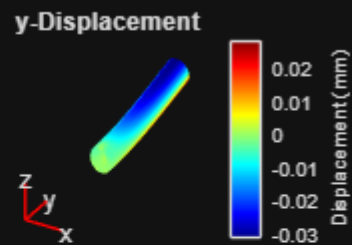
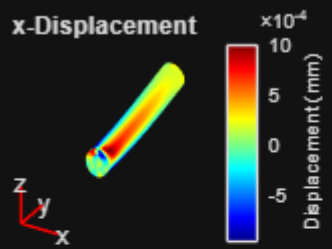
Von Mises Stress

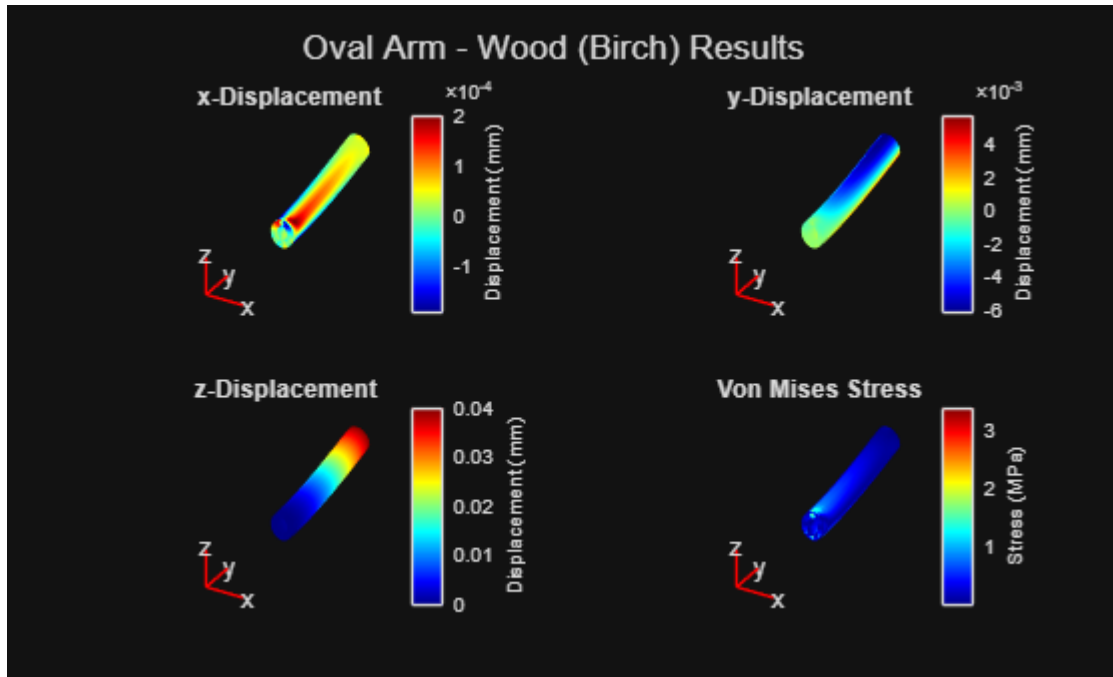


Oval Arm - PLA Plastic Results



Oval Arm - ABS Plastic Results





% Generates final results table

```
designNameColumn = vertcat(results.DesignName);
materialNameColumn = vertcat(results.MaterialName);
maxDisplacementColumn = vertcat(results.MaximumDisplacement) * 1000;
maxVonMisesStressColumn_MPa = vertcat(results.MaximumVonMisesStress) ./ 1e6;
FOSColumn = vertcat(results.FOS);
meetsFOSColumn = FOSColumn >= minFOS;
maxPayloadColumn = reshape(maxPayload, [], 1);
meetsPayloadColumn = reshape(meetsRequirementMinPayload, [], 1);

finalResults = table( ...
    designNameColumn, ...
    materialNameColumn, ...
    maxDisplacementColumn, ...
    maxVonMisesStressColumn_MPa, ...
    FOSColumn, ...
    meetsFOSColumn, ...
    maxPayloadColumn, ...
    meetsPayloadColumn);
finalResults.Properties.VariableNames = ["Design" "Material" "Max Displacement (mm)" "Max Von Mises Stress (MPa)" "FOS" "Meets FOS Requirement?" "Max Payload at TWR = 2 (kg)" "Meets Payload Requirement?"];
```

% Creates uifigure to display results table

```
resultsFig = uifigure( ...
    'Name', 'Design Comparison Results', ...
    'Position', [100 100 1450 600]);
```

```
resultsTable = uitable(resultsFig, ...
    'Data', finalResults, ...
```

```
'Position', [20 20 1410 550]);  
% Exports the image to pre-selected directory  
exportapp(resultsFig, "images/FEA_Results_Table.png");
```

Design	Material	Max Displacement (mm)	Max Von Mises Stress (MPa)	FOS	Meets FOS Requirement?	Max Payload at TWR = 2 (kg)	Meets Payload Requirement?
Truss Arm	Carbon Fiber Composite (CFRP)	0.0224	16.2093	37.0158	☑	0.8973	☑
Truss Arm	Aluminum Alloy	0.0225	15.7667	17.4418	☑	0.8266	☑
Truss Arm	Fiberglass Composite (GFRP)	0.0625	16.3430	18.3565	☑	0.8780	☑
Truss Arm	PLA Plastic	0.4498	15.6747	3.8278	☑	0.9197	☑
Truss Arm	ABS Plastic	0.7887	15.8085	2.5303	☑	0.9326	☑
Truss Arm	Wood (Birch)	0.1583	15.8737	5.0398	☑	0.9615	☑
Oval Arm	Carbon Fiber Composite (CFRP)	0.0057	3.5144	170.7269	☑	0.7928	☑
Oval Arm	Aluminum Alloy	0.0057	3.3609	81.8224	☑	0.6504	☑
Oval Arm	Fiberglass Composite (GFRP)	0.0160	3.5444	84.6409	☑	0.7540	☑
Oval Arm	PLA Plastic	0.1154	3.3548	17.8851	☑	0.8382	☑
Oval Arm	ABS Plastic	0.2025	3.4022	11.7571	☑	0.8641	☑
Oval Arm	Wood (Birch)	0.0408	3.4303	23.3214	☑	0.9223	☑

Interpretation of Results - Final Recommended Design Solution

Using the results of your thrust-to-weight analysis and finite element analysis, recommend a final drone arm design that meets all required constraints. Your recommendation should clearly identify the selected design (arm geometry and material choice), make use of quantitative results to justify your decisions, demonstrate that the design meets the constraints, explain the trade-offs considered, and include comparisons to alternative designs.

Approach and Methodology:

Two drone arms are proposed as a possible solution to this project, a planar truss design and a hollowed oval design. Each design and material combination was first checked to ensure that it could meet the minimum thrust-to-weight ratio (TWR) of 2:1 while also allowing for a minimum payload of 0.5 kg.

Finite element analysis (FEA) was then performed on each combination of to determine maximum displacement, maximum Von Mises stress, and factor of safety (FOS). The minimum requirements for this project are a FOS of 1.5 and a payload capacity of 0.5 kg.

Results and Analysis:

After running the FEA simulation on every combination of arm design and material, all combinations satisfied both the payload and factor-of-safety requirements.

The design with the lowest payload capacity was still able to support approximately 0.6504 kg, which is 30.0% greater than the required minimum payload of 0.500 kg. Likewise, every design exceeded the minimum required FOS of 1.5, with the lowest calculated FOS being approximately 2.53.

Because every design satisfied the minimum requirements, the final recommendation was based on which combination provided the best overall balance of payload capacity, structural stiffness, arm mass, and safety margin.

The standout result was the planar truss design made from carbon fiber composite. It supported a maximum payload of 0.8973 kg while maintaining excellent structural stiffness, with a maximum displacement of only 0.0224 mm. Although the PLA plastic, ABS plastic, and wood versions of the planar truss supported slightly higher maximum payloads, they experienced greater displacement. In contrast, the hollow oval arm designs generally exhibited greater stiffness, but their higher mass resulted in lower payload capacities. This indicates that the carbon fiber truss provides a strong overall balance between payload capability, structural rigidity, and mass efficiency.

Challenges Encountered:

The biggest challenge in this project was defining the boundary conditions for the FEA model. The imported CAD geometries contained many individual faces, which made it difficult to correctly identify the fixed and loaded surfaces. As a result, multiple simulations had to be performed to verify that the selected boundary conditions produced realistic displacement and stress behavior.

Determining the surface area of the selected loaded faces was also challenging. There was no straightforward method in our workflow to automatically calculate the surface area of the specific selected faces. To address this, the required face areas were obtained from the CAD software and then manually incorporated into the MATLAB code for the traction calculations.

Design Decisions and Trade-offs:

The final design selection required balancing several competing factors, including payload capacity, structural stiffness, mass, and factor of safety. No single design performed best in every category, so the final recommendation was based on overall performance rather than one isolated metric.

The planar truss geometry was favored because it reduced unnecessary material while maintaining a strong load path between the motor mount and the drone body. This lower mass improved payload capacity compared with the hollow oval design. However, the hollow oval arms generally exhibited lower displacement, indicating greater stiffness under the same loading conditions.

Material selection involved a similar trade-off. PLA plastic, ABS plastic, and wood allowed the planar truss to carry slightly greater payloads than carbon fiber because of their lower densities. However, those materials also experienced larger displacements. Carbon fiber composite provided a stronger balance between low mass and stiffness, supporting a payload of 0.8973 kg while limiting maximum displacement to only 0.0224 mm.

Although some alternative designs achieved either a higher payload capacity or lower displacement, the carbon fiber planar truss performed well in both categories while also exceeding the minimum factor-of-safety requirement of 1.5. For this reason, the carbon fiber composite planar truss was selected as the final

recommended design because it provides the best overall compromise between payload capacity, structural rigidity, mass efficiency, and safety.